

# WILL Part I (addition)

## *Closed Algebraic System of Relational Orbital Mechanics (R.O.M.)*

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October 2025

## 1 Closed Algebraical System of R.O.M. Equations

Dedicated sections in WILL RG Part I: [https://willrg.com/documents/WILL RG I.pdf-sec:rom](https://willrg.com/documents/WILL_RG_I.pdf-sec:rom)

Desmos Project

**R.O.M.**

**Remark 1.1.** *R.O.M. does not describe how a body moves under forces; it classifies the algebraically allowed relational states of a bound system. It is not equations of motion but algebraically closed system of allowed states within WILL - allowed free will.*

$$\kappa^2 = 1 - \frac{1}{(1 + z_\kappa)^2} \quad (z_\kappa = \text{gravitational redshift})$$

$$\beta^2 = 1 - \frac{1}{(1 + z_\beta)^2} \quad (z_\beta = \text{transverse Doppler shift})$$

### Observational Inputs

$Z_{sys} = (1 + z_\kappa)(1 + z_\beta) = \frac{1}{\tau}$  (product of gravitational red shift and transverse Doppler shift)

$\tau = \kappa_X \cdot \beta_Y = \frac{1}{Z_{sys}}$  (product of projectinal phase factors on  $S^1$  and  $S^2$  carriers)

$z_\kappa = \frac{1}{\kappa_X} - 1$  (gravitational redshift)

$z_\beta = \frac{1}{\beta_Y} - 1$  (transverse Doppler shift)

$z_{\kappa o} = \frac{1}{\kappa_{Xo}} - 1$  (redshift at phase  $o$ )

$z_{\beta o} = \frac{1}{\beta_{Yo}} - 1$  (transverse Doppler shift at phase  $o$ )

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\*This work is archived on Zenodo: <https://zenodo.org/record/17115270>

## Global System Parameters (Fixed for the Orbit)

$$\begin{aligned}
\kappa &= \sqrt{\frac{R_s}{a}} = \sqrt{\frac{\rho}{\rho_{max}}} = \sqrt{\kappa_p^2(1-e)} = \sqrt{4W} = \sqrt{2(\kappa_o(o)^2 - \beta_o(o)^2)} = \sqrt{\frac{1}{2}(3 - \sqrt{1 + 8\tau_{Wo}(O_o)^2})} = \\
&\sqrt{1 - (1 + z_\kappa)^{-2}} = \kappa_R^2 \left(\frac{I_t}{T}\right)^{\frac{2}{3}} = \kappa_o^2 \cdot \eta_o \text{ (potential projection at semi-major axis)} \\
\kappa_X &= \cos(\theta_2) = \sqrt{1 - \kappa^2} = \frac{1}{(1+z_\kappa)} \text{ (potential phase factor)} \\
\beta &= \frac{\kappa}{\sqrt{2}} = \beta_p \sqrt{\frac{1-e}{1+e}} = \sqrt{2W} = \frac{4\pi^2 a^2}{T^2 c^2} = \left(\frac{R_s}{\frac{T}{\pi} c}\right)^{\frac{2}{3}} = \sqrt{(\kappa_o(o)^2 - \beta_o(o)^2)} = \sqrt{\frac{\kappa_p^2}{2}(1-e)} = \\
&= \sqrt{\kappa_o^2 - \frac{\kappa_o^2}{2} \cdot \left(1 + \left(\frac{1}{\delta_o(o)} - 1\right)\right)} = \beta_o(o) \frac{\sqrt{1-e^2}}{\sqrt{1+e^2+2e \cdot \cos(o)}} = \sqrt{1 - (1 + z_\beta)^{-2}} \text{ (kinetic pro-} \\
&\text{jection on semi major axis)} \\
\beta_Y &= \sin(\theta_1) = \sqrt{1 - \beta^2} \text{ (kinetic phase factor)} \\
\beta_{int} &= \frac{\beta}{\sqrt{1-e^2}} \text{ (interior kinetic projection)} \\
\tau &= \kappa_X \beta_Y \text{ (relational spacetime factor)} \\
\tau_Y &= \sqrt{1 - \tau^2} = \sqrt{3\beta^2 - 2\beta^4} \text{ (relational spacetime phase factor)} \\
Q &= \sqrt{\kappa^2 + \beta^2} = \sqrt{\frac{3}{2}} \kappa = \sqrt{3} \beta \text{ (total relational shift (magnitude of state difference))} \\
R_s &= \kappa_o(o)^2 r_o(o) = \frac{2Gm_0}{c^2} = \frac{\Delta t_o(o)}{\zeta(o)} \kappa^2 \beta c = Tc \frac{\beta^3}{\pi} = t_o(o) \kappa_o(o)^2 \cdot c = \frac{r_1 r_2}{r_2 - r_1} (\beta_1^2 - \beta_2^2) = \\
&\frac{a}{2} (3 - \sqrt{1 + 8\tau_{Wo}(O_o)^2}) = \\
&= \frac{r_o(o)}{2(2a - r_o(o))} (4a - r_o(o) - \sqrt{(4a - r_o(o))^2 - 8a(2a - r_o(o))(1 - \tau_{Wo}(o)^2)}) = \frac{T}{\pi} \left(\sqrt{1 - \left(\frac{1}{1+z_\beta}\right)^2}\right)^3 c \\
&\text{(Schwarzschild radius - system scale)} \\
a &= \frac{R_s}{\kappa^2} = \frac{R_s}{4W} = \frac{\beta_o c}{\omega} \cdot \frac{\sqrt{1-e^2}}{\sqrt{1+e^2+2e \cos(o)}} = \frac{\Delta t_o(o)}{\zeta} \beta c \text{ (semi-major axis)} \\
m_0 &= \frac{\kappa^2 c^2 a}{2G} = 4\pi \rho a^3 \text{ (mass parameter)} \\
t &= \frac{a}{c} \text{ (temporal scale of the system)} \\
W &= \frac{\beta^2}{2} = \frac{1}{2}(\kappa_o^2 - \beta_o^2) = \frac{1}{4}\kappa_p^2(1-e) = \frac{1}{2}(\kappa_o(o)^2 - \frac{\kappa_o(o)^2}{2\delta_o(o)}) = \frac{1}{2}((1 - (1 + z_{\kappa_o}(o))^{-2}) - (1 - \\
&(1 + z_{\beta_o}(o))^{-2})) = \left(\frac{\pi R_s}{2\sqrt{2}cT_o}\right)^{\frac{2}{3}} \text{ (energy invariant - binding energy)} \\
\Delta\phi &= \frac{2\pi \cdot \tau_Y^2}{1-e^2} = \frac{2\pi(3\beta^2 - 2\beta^4)}{1-e^2} \text{ (precession of perihelion per orbit)} \\
h_W &= r_o(o) \beta_T(o) c = r_o^2 \cdot \omega_o = a \cdot \beta c \cdot e_Y = R_s c \frac{\sqrt{1-e^2}}{2\beta} \text{ (invariant angular momentum)} \\
\omega &= \frac{\beta c}{a} \text{ (angular frequency)} \\
T &= \frac{2\pi}{\omega} = \frac{2\pi a}{\beta c} = \frac{2\pi r_o(O_o)}{\beta_o(O_o) c} = \frac{\pi R_s}{c\beta^3} = \frac{\pi R_s}{2\sqrt{2}c} W^{-\frac{3}{2}} \text{ (orbital period)}
\end{aligned}$$

## Eccentricity Relations

$$\begin{aligned}
e &= \frac{1}{\delta_p} - 1 = 1 - \frac{2\beta_a^2}{\kappa_a^2} = \frac{2\beta_p^2}{\kappa_p^2} - 1 \text{ (eccentricity derived from closure)} \\
e_Y &= \sqrt{1 - e^2} \text{ (eccentricity's orthogonal value)} \\
e_X &= \frac{1+e}{1-e} = \frac{r_a}{r_p} = \frac{\delta_a}{\delta_p} = \frac{\beta_p}{\beta_a} = \frac{\kappa_p^2}{\kappa_a^2} = \frac{\kappa_a^2 \beta_p^2}{\kappa_p^2 \beta_a^2} \text{ (shape factor, still looks like magic to me)}
\end{aligned}$$

## Time-phase

$$\begin{aligned}
\omega_o(o) &= \frac{\beta \cdot c}{a} \cdot \frac{(1+e \cdot \cos(o))^2}{(1-e^2)^{\frac{3}{2}}} \text{ (angular frequency at phase o)} \\
\Delta t_o(o) &= \int_0^o \frac{1}{\omega_\theta(\theta)} d\theta = \frac{a}{\beta c} \zeta = \frac{T_o}{2\pi} \zeta \text{ (time duration of given phase interval)} \\
\zeta &= (1 - e^2)^{\frac{3}{2}} \cdot \left(\int_0^o (1 + e \cdot \cos(\theta))^{-2} d\theta\right) = \frac{1}{\sqrt{1-e^2}} \int_0^o \left(\frac{t_o(\theta)}{t}\right)^2 d\theta \text{ (Temporal Phase interval)}
\end{aligned}$$

## Perihelion Relations

$$\begin{aligned}
 r_p &= a(1 - e) = \frac{R_s}{\kappa_p^2} \text{ (radius at perihelion)} \\
 \kappa_p &= \kappa \sqrt{\frac{1}{1-e}} = \sqrt{\frac{2\beta^2}{1-e}} = Q_p \sqrt{\frac{2}{3+e}} \text{ (potential projection at perihelion)} \\
 \kappa_{Xp} &= \sqrt{1 - \kappa_p^2} \text{ (potential phase projection at perihelion)} \quad \beta_p = \frac{V_p}{c} = \beta \sqrt{\frac{1+e}{1-e}} = \frac{\kappa_p}{\sqrt{2\delta_p}} = \\
 &\frac{\kappa_p}{\sqrt{2}} \sqrt{1+e} \text{ (kinetic projection at perihelion)} \\
 \delta_p &= \frac{\kappa_p^2}{2\beta_p^2} = \frac{1}{1+e} \text{ (closure factor at perihelion)} \\
 Q_p &= \sqrt{\kappa_p^2 + \beta_p^2} \text{ (relational shift at perihelion)}
 \end{aligned}$$

## Aphelion Relations

$$\begin{aligned}
 r_a &= a(1 + e) = \frac{R_s}{\kappa_a^2} \text{ (radius at aphelion)} \\
 \beta_a &= \sqrt{\beta_p^2 e_X^{-2}} = \beta \sqrt{e_X^{-1}} \text{ (kinetic projection at aphelion)} \\
 \kappa_a &= \sqrt{2W + \beta_a^2} \text{ (potential projection at aphelion)} \\
 \delta_a &= \frac{1}{1-e} = \frac{\kappa_a^2}{2\beta_a^2} \text{ (closure factor at aphelion)} \\
 Q_a &= \sqrt{\kappa_a^2 + \beta_a^2} \text{ (relational shift at aphelion)}
 \end{aligned}$$

## Phase Variables (Depend on $o$ )

$o$  = orbital phase in radians

$$\begin{aligned}
 r &= r_o(o) = a \frac{1-e^2}{1+e \cos o} = \frac{R_s}{\kappa_o^2} \text{ (radial distance at phase } o) \\
 \kappa_o &= \sqrt{\beta_R(o)^2 + \beta_T(o)^2 + \beta^2} = \sqrt{\frac{R_s}{r}} = 1 - (1 + z_{\kappa o}(o))^{-2} = \kappa_p \sqrt{\frac{1+e \cos(o)}{1+e}} = \beta^2 + \beta_o^2 = \\
 &\sqrt{\kappa_{surface}^2 \cdot \sin(\theta_{obs})} \text{ (local potential at phase } o) \\
 \kappa_{Xo} &= \sqrt{1 - \kappa_o^2} = (z_{\kappa o}(o) + 1)^{-1} \text{ (gravitational phase factor at phase } o) \\
 \beta_o(o) &= \beta \cdot \frac{\sqrt{1+e^2+2 \cdot e \cdot \cos(o)}}{\sqrt{1-e^2}} = \sqrt{\kappa_o^2 - 2W} = \frac{\kappa_o(o)}{\sqrt{2\delta_o(o)}} = \sqrt{\frac{\kappa_o(o)^2}{2} \frac{1+e^2+2 \cdot e \cdot \cos(o)}{1+e \cdot \cos(o)}} = \sqrt{\frac{R_s}{r_o(o)} - \frac{R_s}{2a}} = \\
 &\sqrt{1 - (1 + z_{\beta o}(o))^{-2}} \text{ (kinetic projection at phase } o) \\
 \beta_R(o) &= \beta \frac{e \sin(o)}{\sqrt{1-e^2}} = \sqrt{\beta_o(o)^2 - \beta_T(o)^2} = \sqrt{((1 - (1 + z_{\kappa o}(o))^{-2}) - 2W) - \frac{r_o(o)^2 \omega_o(o)^2}{c^2}} \text{ (ra-} \\
 &\text{dial kinetic projection at phase } o) \\
 \beta_T(o) &= \frac{r_o(o) \omega_o(o)}{c} = \beta \frac{1+e \cos(o)}{\sqrt{1-e^2}} = \kappa_o^2 \frac{\sqrt{1-e^2}}{2\beta} \text{ (transverse kinetic projection at phase } o) \\
 \beta_{Yo} &= \sqrt{1 - \beta_o^2} = (z_{\beta o}(o) + 1)^{-1} \text{ (relativistic phase factor at phase } o) \\
 \delta_o &= \frac{1+e \cos o}{1+e^2+2e \cos o} = \frac{\kappa_o^2}{2\beta_o^2} \text{ (local closure factor at phase } o) \\
 Q_o &= \sqrt{\kappa_o^2 + \beta_o^2} \text{ (local relational shift vector at phase } o) \\
 \omega_o &= a\beta c \frac{e_Y}{r_o^2} = \frac{\beta c}{a} \frac{(1+e \cos o)^2}{(1-e^2)^{3/2}} \text{ (angular speed at phase } o) \\
 \eta_o &= \frac{r}{a} = 2 - \frac{2\beta_o(o)^2}{\kappa_o(o)^2} \text{ (phase scale amplitude at phase } o) \\
 \tau(o) &= \kappa_{Xo}(o) \cdot \beta_{Yo}(o) = Z_{sys}(o)^{-1} \text{ (phase spacetime factor at phase } o) \\
 t_o &= \frac{r_o(o)}{c} \text{ (light time scale at phase } o)
 \end{aligned}$$

$$\Delta_o = 3\beta_{int}^2 \cdot o \text{ (precession of perihelion at phase } o)$$

## Orbital Phase Invariants

### The Decryption Invariant

$$Z_{raw}(-\omega_i)\tau(-\omega_i)(1 - e \cos \omega_i) + Z_{raw}(\pi - \omega_i)\tau(\pi - \omega_i)(1 + e \cos \omega_i) = 2 \quad (1)$$

$$\begin{aligned} W &= \frac{1}{2}(\kappa_o^2 - \beta_o^2) \\ h_W &= r_o^2 \cdot \omega_o \\ \frac{\beta_T(o)}{\kappa_o^2(o)} &= \frac{\sqrt{1-e^2}}{2\beta} \\ \beta_T^2 \eta_o^2 &= \beta^2(1 - e^2) \end{aligned}$$

## Relational Geometry (WILL)

$$\begin{aligned} \theta_1 &= \arccos(\beta) \text{ (distribution angle on } S^1, \text{ non-physical)} \\ \theta_2 &= \arcsin(\kappa) \text{ (distribution angle on } S^2, \text{ non-physical)} \\ \Delta_Q &= Q_o^2 - Q^2 \text{ (phase relational shift amplitude at phase } o) \\ O_o &= \arccos(1 - \delta^{-1}) = \arccos(-e) = \arccos(1 - \frac{2\beta_p^2}{\kappa_p^2}) = \arccos(\frac{2\beta_a^2}{\kappa_a^2} - 1) \text{ (orbital balance} \\ &\text{point where } \kappa_o^2 = 2\beta_o^2 \text{ is true)} \end{aligned}$$

## Structural-Dynamical Equivalence

$$\frac{r_a}{r_p} = \frac{\delta_a}{\delta_p} = \frac{1+e}{1-e} = \frac{\beta_p}{\beta_a} = \frac{\kappa_p^2}{\kappa_a^2} = \frac{\kappa_a^2 \beta_p^2}{\kappa_p^2 \beta_a^2} \text{ (remarkable equivalence of structural } (\kappa \text{ on } S^2 \text{ carrier)}$$

and dynamical ( $\beta$  on  $S^1$  carrier) symmetry suggesting once again that  $SPACETIME \equiv ENERGY$ )

## Observer dependant

$$\begin{aligned} Z_{raw}(o) &= (1 + \beta_{int}(\cos(o + \omega_i) + e \cos(\omega_i)) \sin(i)) Z_{sys}(o) \text{ (raw light shift including the} \\ &\text{line of site Doppler at phase } o) \\ \beta_{los}(o) &= \frac{\beta}{\sqrt{1-e^2}}(\cos(o + \omega_i) + e \cos(\omega_i)) \sin(i) \text{ (line of site light shift)} \\ i &= \text{(orbital inclination in relation to line of site and orbital plane)} \\ \omega_i &= \text{(phase turn or argument of periapsis)} \\ K_i &= \frac{\beta}{\sqrt{1-e^2}} \sin(i) = \beta_{int} \sin(i) \text{ (semi-amplitude invariant)} \\ Z_{rawmax} &= Z_{sys}(-\omega_i)(1 + K_i(1 + e \cos \omega_i)) \\ Z_{rawmin} &= Z_{sys}(\pi - \omega_i)(1 + K_i(-1 + e \cos \omega_i)) \\ O_{oi} &= (O_o + \omega_i) \\ Z_{sys}(-\omega_i) &= (1 - \beta^2 \frac{1+e^2+2e \cos(\omega_i)}{1-e^2})^{-\frac{1}{2}} (1 - 2\beta^2 \frac{1+e \cos(\omega_i)}{1-e^2})^{-\frac{1}{2}} \\ Z_{sys}(\pi - \omega_i) &= (1 - \beta^2 \frac{1+e^2-2e \cos(\omega_i)}{1-e^2})^{-\frac{1}{2}} (1 - 2\beta^2 \frac{1-e \cos(\omega_i)}{1-e^2})^{-\frac{1}{2}} \end{aligned}$$

## Background Free Parametric Equations for the Observed Orbit

$$\begin{bmatrix} x_{sky} \\ y_{sky} \\ z_{depth} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(i) & -\sin(i) \\ 0 & \sin(i) & \cos(i) \end{bmatrix} \begin{bmatrix} x_{orb} \\ y_{orb} \\ 0 \end{bmatrix}$$

$$\begin{aligned} x_{sky} &= r(o) \cos(o + \omega_i) \\ y_{sky} &= r(o) \sin(o + \omega_i) \cos(i) \\ z_{depth} &= r(o) \sin(o + \omega_i) \sin(i) \end{aligned}$$

## Un-tilted 2D coordinates within the orbital plane

$$\begin{aligned} x_{orb} &= r(o) \cos(o + \omega_i) \\ y_{orb} &= r(o) \sin(o + \omega_i) \end{aligned}$$